Stick-together calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a common final velocity using total momentum.

SUCCESS CRITERIA - I can...

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two carts separate from rest with no external horizontal impulse. One has momentum 15 kg·m/s left. Find the other's momentum.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. v_f is the shared final velocity.

Worked reasoning: A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

Your turn: Mark the conserved quantity on a before-and-after sketch. Explain why masses are added in the final denominator.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two carts stick. Identify which masses belong in the final momentum expression and why.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A 1 kg cart moves right at 6 m/s and sticks to a 2 kg cart at rest. Find initial total momentum, combined mass and final velocity. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

Stick-together calculations

antocciphysics

Continue down each column.

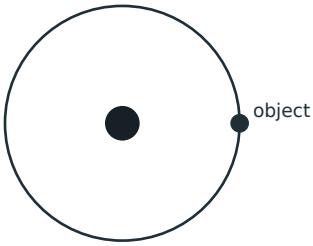
05 Explain your reasoning

A student averages the two starting velocities without considering mass. When would that happen to work?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A car follows a circle at steady 6 m/s. Sketch velocity at two points. Could net force be zero while it follows this path?
Label velocity and inward acceleration on the circular model.



Marzano check - circle: 1 2 3

07 Exit check

Two equal 1 kg carts approach at 2 m/s in opposite directions and stick. Find final velocity. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose the mass of a stationary cart so a 2 kg cart at 6 m/s slows to 2 m/s when they stick. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3